

Aswin Kumar B S

LinkedIn - <https://www.linkedin.com/in/aswinkumar7/>

Email: aswin.aswin5640@gmail.com

GitHub - <https://github.com/Aswin-Kumar7>

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, Java, C
- **Frontend:** React.js, Next.js
- **Backend & Frameworks:** Node.js, Express.js, Flask
- **Database & Tools:** MongoDB, Postgres, Veeam Cloud Replication
- **Cloud & DevOps:** AWS, Azure, CI/CD Pipelines, Docker, Kubernetes, Veeam Backup and Replication
- **AI & ML:** TensorFlow, TensorFlow Lite, Deep Learning, On-device ML, Scikit-learn, LLMs, RAG, LangChain
- **Version Control:** Git, GitHub
- **AI Tools:** Claude, Antigravity, Cursor, Codex

PROJECTS

• RoadSoS — AI Road Accident Emergency Response System

Real-time crash detection and emergency dispatch system powered by an on-device classifier trained on the VZCrash IMU dataset (~190K events) reaching 0.9948 ROC-AUC, deployed via TensorFlow Lite on Android for offline inference, with offline SMS alerts and a live dispatcher dashboard. Handles 3-class classification (crash / near-miss / normal driving) on 100 Hz accelerometer + gyroscope time-series data. (TFLite, Python, Flutter, FastAPI, Vue 3.)

• Real-Time Call Center Copilot System | Github

Built a low-latency AI copilot for **real-time transcription**, **sentiment analysis**, and **response generation** using Web Sockets, FastAPI, Twilio, and LLaMA (Groq) with Pinecone-based RAG for contextual insights and summaries.

• Fake News Detection System | Github

Developed a real-time fake news detection system using Gemini AI and SerpAPI for **text/image verification** with a multi-language responsive UI. Tools: JavaScript (Manifest V3), FastAPI, Scikit-learn.

• NGO Pet Adoption System | Github

Engineered a **MERN-based** adoption platform with PhonePe payment integration and **AI-powered pet matching** using HuggingFace embeddings, LangChain, and Pinecone vector search.

ACHIEVEMENTS

- **Winner — BIMSTEC AI Road Safety Hackathon 2026, IIT Madras (CoERS).** Ranked **1st** of 19,000+ registrations / 6,200+ submissions (**1 of 21 finalists**) for RoadSoS, an offline-first AI accident detection and emergency-dispatch system built to save lives in the "golden hour."
- **Winner – Innovate & Build 2025 Hackathon** - Secured **1st place** in a **two-phase hackathon** by demonstrating strong innovation, problem-solving, and execution in the final round.
- **Runner-Up – INFYND 36-Hour AI Hackathon (AI, Full-Stack, RAG Systems)** Achieved **2nd place** by developing Aura AI, a **real-time call center copilot** using LangChain, Groq (LLaMA), Pinecone, and Next.js, showcasing rapid prototyping and scalable AI system design under time constraints.
- **Finalist at Cognizant Technoverse 2026**, recognized among the **top 100 participants** in a nation-level hackathon for developing an innovative technology solution using **GenAI**

EXPERIENCE

- **Full Stack Development Intern, Erthaloka (Remote)** Nov 2025
Developed **MERN-based full-stack applications** with **RESTful APIs**, database integration, and responsive UI, implementing Google OAuth, secure authentication flows, payment gateway integration, and TypeScript-based scalable architecture.
- **Cloud Computing Intern, UBX Cloud (On-site)** May 2023
Configured and managed **Veeam Backup & Replication** and **VSPC**, handling backup policies, agent configuration, monitoring, and disaster recovery to ensure high availability and data protection.
- **Community Manager, Information Data Systems (Remote)** May 2023
Scaled and managed Discord, LinkedIn, and Instagram communities, increasing engagement by 40% through growth strategies, content planning, and event-driven user engagement.

RESEARCH PAPER

Optimized UAV Trajectory Planning for Precision Agriculture Using Wireless Sensor Networks (IEEE Conference, ICAECA 2025)

- Proposed an optimized UAV trajectory planning approach for precision agriculture using wireless sensor networks, contributing to system modeling, optimization, and performance evaluation in smart agriculture applications.

EDUCATION

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| • Kumaraguru College of Technology, Coimbatore | 2024 – 2027 |
| B.E Computer Science and Engineering | |
| • PSG Polytechnic College, Coimbatore | 2021 – 2024 |
| Diploma in Computer Networking | |

CORE INTERESTS

- Analytical & Critical Thinking, Structured Problem Solving
- Technical & Professional Communication
- Team Collaboration & Coordination